

1. Yes the test case 1 has passed successfully., the output for the test case 1 is:

```
[0] checkpoint successful ao: 0
[1] checkpoint successful ao: 0
[2] checkpoint successful ao: 0
[3] checkpoint successful ao: 0
[4] checkpoint successful ao: 0
[5] checkpoint successful ao: 0
[6] checkpoint successful ao: 0
[7] checkpoint successful ao: 0
[8] checkpoint successful ao: 0
[9] checkpoint successful ao: 0
[10] checkpoint successful ao: 0
[11] checkpoint successful ao: 0
[12] checkpoint successful ao: 0
[13] checkpoint successful ao: 0
[14] checkpoint successful ao: 0
[15] checkpoint successful ao: 0
[16] checkpoint successful ao: 0
[17] checkpoint successful ao: 0
[18] checkpoint successful ao: 0
[19] checkpoint successful ao: 0
[20] checkpoint successful ao: 0
[21] checkpoint successful ao: 0
[22] checkpoint successful ao: 0
[23] checkpoint successful ao: 0
[24] checkpoint successful ao: 0
[25] checkpoint successful ao: 0
[26] checkpoint successful ao: 0
[27] checkpoint successful ao: 0
[28] checkpoint successful ao: 0
[29] checkpoint successful ao: 0
[30] checkpoint successful ao: 0
[31] checkpoint successful ao: 0
test PASSED
```

```
iiitd@iiitd-ThinkCentre-M70s-Gen-3:~/Desktop/project2/rdt$ ./sender
[0] checkpoint successful ao: 0
[1] checkpoint successful ao: 0
[2] checkpoint successful ao: 0
[3] checkpoint successful ao: 0
[4] checkpoint successful ao: 0
[5] checkpoint successful ao: 0
[6] checkpoint successful ao: 0
[7] checkpoint successful ao: 0
[8] checkpoint successful ao: 0
[9] checkpoint successful ao: 0
[10] checkpoint successful ao: 0
[11] checkpoint successful ao: 0
[12] checkpoint successful ao: 0
[13] checkpoint successful ao: 0
[14] checkpoint successful ao: 0
[15] checkpoint successful ao: 0
[16] checkpoint successful ao: 0
[17] checkpoint successful ao: 0
[18] checkpoint successful ao: 0
[19] checkpoint successful ao: 0
[20] checkpoint successful ao: 0
[21] checkpoint successful ao: 0
[22] checkpoint successful ao: 0
[23] checkpoint successful ao: 0
[24] checkpoint successful ao: 0
[25] checkpoint successful ao: 0
[26] checkpoint successful ao: 0
[27] checkpoint successful ao: 0
[28] checkpoint successful ao: 0
[29] checkpoint successful ao: 0
[30] checkpoint successful ao: 0
[31] checkpoint successful ao: 0
test PASSED
```

2. Yes the Test Case 2 is passing and the the output for the test case 1 is:

```
[0] checkpoint successful ao: 5
[1] checkpoint successful ao: 5
[2] checkpoint successful ao: 4
[3] checkpoint successful ao: 4
[4] checkpoint successful ao: 3
[5] checkpoint successful ao: 3
[6] checkpoint successful ao: 3
[7] checkpoint successful ao: 4
[8] checkpoint successful ao: 4
[9] checkpoint successful ao: 4
[10] checkpoint successful ao: 4
[11] checkpoint successful ao: 4
[12] checkpoint successful ao: 4
[13] checkpoint successful ao: 4
[14] checkpoint successful ao: 5
[15] checkpoint successful ao: 5
```

[16] checkpoint successful ao: 5
[17] checkpoint successful ao: 4
[18] checkpoint successful ao: 5
[19] checkpoint successful ao: 5
[20] checkpoint successful ao: 5
[21] checkpoint successful ao: 5
[22] checkpoint successful ao: 4
[23] checkpoint successful ao: 4
[24] checkpoint successful ao: 4
[25] checkpoint successful ao: 4
[26] checkpoint successful ao: 4
[27] checkpoint successful ao: 4
[28] checkpoint successful ao: 5
[29] checkpoint successful ao: 5
[30] checkpoint successful ao: 5
[31] checkpoint successful ao: 5
test PASSED

```
liitd@liitd-ThinkCentre-M70s-Gen-3:~/Desktop/project2/rdt$ ./sender 1
[0] checkpoint successful ao: 5
[1] checkpoint successful ao: 5
[2] checkpoint successful ao: 5
[3] checkpoint successful ao: 4
[4] checkpoint successful ao: 4
[5] checkpoint successful ao: 4
[6] checkpoint successful ao: 4
[7] checkpoint successful ao: 4
[8] checkpoint successful ao: 4
[9] checkpoint successful ao: 4
[10] checkpoint successful ao: 4
[11] checkpoint successful ao: 4
[12] checkpoint successful ao: 4
[13] checkpoint successful ao: 4
[14] checkpoint successful ao: 5
[15] checkpoint successful ao: 5
[16] checkpoint successful ao: 4
[17] checkpoint successful ao: 4
[18] checkpoint successful ao: 4
[19] checkpoint successful ao: 4
[20] checkpoint successful ao: 4
[21] checkpoint successful ao: 4
[22] checkpoint successful ao: 4
[23] checkpoint successful ao: 5
[24] checkpoint successful ao: 4
[25] checkpoint successful ao: 4
[26] checkpoint successful ao: 4
[27] checkpoint successful ao: 4
[28] checkpoint successful ao: 5
[29] checkpoint successful ao: 5
[30] checkpoint successful ao: 5
[31] checkpoint successful ao: 5
test PASSED
```

3) Run time in Test 1:

- a.) real: 5.687 seconds
- b.) user: 0.569 seconds
- c.) sys: 0.061 seconds

4) Run time in Test 2:

- a.) real: 10.488 seconds
- b.) user: 1.340 seconds
- c.) sys: 0.057 seconds

5.)

For sender

The value of the maximum resident set size is 2328 kb and the value of Elapsed (wall clock) time is 40.65 seconds ,for sender we have obtained the following values

```
Command being timed: "./sender 1"
User time (seconds): 0.39
System time (seconds): 0.12
Percent of CPU this job got: 1%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:40.65
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 2328
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 0
Minor (reclaiming a frame) page faults: 2234
Voluntary context switches: 30009
Involuntary context switches: 9
Swaps: 0
File system inputs: 0
File system outputs: 0
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

For Receiver:

The value of the maximum resident set size is 1269468 kb and the value of Elapsed (wall clock) time is 10.29 seconds ,for sender we have obtained the following values

```
Command being timed: "./receiver"
User time (seconds): 0.50
System time (seconds): 0.56
Percent of CPU this job got: 10%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:10.29
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 1269468
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 0
Minor (reclaiming a frame) page faults: 322998
Voluntary context switches: 2750
Involuntary context switches: 42
Swaps: 0
File system inputs: 0
File system outputs: 0
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

6) For sender

The value of the maximum resident set size is 1760 kb and the value of Elapsed (wall clock) time is 7.73 seconds ,for sender we have obtained the following values

```
[22] checkpoint successful ao: 0
[23] checkpoint successful ao: 0
[24] checkpoint successful ao: 0
[25] checkpoint successful ao: 0
[26] checkpoint successful ao: 0
[27] checkpoint successful ao: 0
[28] checkpoint successful ao: 0
[29] checkpoint successful ao: 0
[30] checkpoint successful ao: 0
[31] checkpoint successful ao: 0
test PASSED
    Command being timed: "./sender"
    User time (seconds): 0.51
    System time (seconds): 0.04
    Percent of CPU this job got: 7%
    Elapsed (wall clock) time (h:mm:ss or m:ss): 0:07.73
    Average shared text size (kbytes): 0
    Average unshared data size (kbytes): 0
    Average stack size (kbytes): 0
    Average total size (kbytes): 0
    Maximum resident set size (kbytes): 1760
    Average resident set size (kbytes): 0
    Major (requiring I/O) page faults: 0
    Minor (reclaiming a frame) page faults: 165
    Voluntary context switches: 4197
    Involuntary context switches: 5
    Swaps: 0
    File system inputs: 0
    File system outputs: 0
    Socket messages sent: 0
    Socket messages received: 0
    Signals delivered: 0
    Page size (bytes): 4096
    Exit status: 0
```

For receiver

The value of the maximum resident size is 1196640 kb and the value of time elapsed(wall clock) is 4.41 seconds. and The values we have got at the receiver are as follows:

```
liitd@liitd-ThinkCentre-M70s-Gen-3:~/Desktop/project2/rdt$ /usr/bin/time -v ./receiver
Command being timed: "./receiver"
User time (seconds): 0.20
System time (seconds): 0.39
Percent of CPU this job got: 13%
Elapsed (wall clock) time (h:mm:ss or m:ss): 0:04.41
Average shared text size (kbytes): 0
Average unshared data size (kbytes): 0
Average stack size (kbytes): 0
Average total size (kbytes): 0
Maximum resident set size (kbytes): 1196640
Average resident set size (kbytes): 0
Major (requiring I/O) page faults: 0
Minor (reclaiming a frame) page faults: 299024
Voluntary context switches: 1441
Involuntary context switches: 14
Swaps: 0
File system inputs: 0
File system outputs: 0
Socket messages sent: 0
Socket messages received: 0
Signals delivered: 0
Page size (bytes): 4096
Exit status: 0
```

Team Members-

- 1) Aakash Chityala (2023005)
- 2) Akshat Verma (2023065)
- 3) Ansh Gaur (2023098)
- 4) Bhavik Garg (2023170)
- 5) Vidush Jindal (2023592)